An Exact Prime-Square Residue Classification

The first prime-index supercongruence of OEIS A004982

Primary author (AI): ChatGPT (OpenAI)

14 November 2026

Abstract. OEIS A004982 conjectures that $a(pn) \equiv a(n) \pmod{p^2}$ when $p \equiv 1 \pmod 4$. We prove the $n = 1$ case and the complementary classification for all odd primes: $a(p) \equiv 6 \pmod{p^2}$ for $p \equiv 1 \pmod 4$ and $a(p) \equiv 2 \pmod{p^2}$ for $p \equiv 3 \pmod 4$. In particular the restriction on prime residue classes is necessary already at $n = 1$. This is a partial result; the general conjecture is not settled here.

The conjecture and our theorem

The sequence A004982 is

$$a(n) = \frac{2^n}{n!} \prod_{k=0}^{n-1} (4k+3), \qquad a(0) = 1, \quad a(1) = 6.$$

Its OEIS entry conjectures $a(pn) \equiv a(n) \pmod{p^2}$ for every positive n and every prime $p \equiv 1 \pmod 4$.

Theorem 1. *For every odd prime p,*

$$\boxed{a(p) \equiv \begin{cases} 6 \pmod{p^2}, & p \equiv 1 \pmod 4, \\ 2 \pmod{p^2}, & p \equiv 3 \pmod 4. \end{cases}}$$

Consequently $a(p) \equiv a(1) \pmod{p^2}$ holds for odd primes precisely when $p \equiv 1 \pmod 4$.

A harmonic-sum lemma

Write $H_j = \sum_{i=1}^{j} 1/i$ (with $H_0 = 0$), interpreted modulo p when $j < p$, and put $q = (2^{p-1} - 1)/p$.

Lemma 1. *For every odd prime p, the following hold modulo p:*

$$H_{p-1} = 0, \qquad H_{(p-1)/2} = -2q, \qquad H_{\lfloor p/4 \rfloor} = -3q.$$

Also $H_{p-1-j} = H_j$ for $0 \le j \le p-1$.

Proof. Pairing i with $p - i$ proves $H_{p-1} \equiv 0$. The binomial expansion and $\binom{p}{k} \equiv p(-1)^{k-1}/k \pmod{p^2}$ give

$$2q = \frac{2^p - 2}{p} \equiv \sum_{k=1}^{p-1} \frac{(-1)^{k-1}}{k} = H_{p-1} - H_{(p-1)/2} \equiv -H_{(p-1)/2}.$$

Reflecting k to $p - k$ in the alternating sum shows that the same sum is twice its first half modulo p. Hence

$$q \equiv \sum_{k=1}^{(p-1)/2} \frac{(-1)^{k-1}}{k} = H_{(p-1)/2} - H_{\lfloor p/4 \rfloor}.$$

Together these identities imply $H_{\lfloor p/4 \rfloor} \equiv -3q$. Finally, removing the last j terms from H_{p-1} and using $(p-i)^{-1} \equiv -i^{-1}$ gives $H_{p-1-j} \equiv H_j$. □

Prime-square computation

Proof of Theorem 1. Let $r \in \{1,3\}$ be chosen so that $rp \equiv 3 \pmod 4$, and set $j = (rp-3)/4$. Thus $r = 3$ if $p \equiv 1 \pmod 4$ and $r = 1$ if $p \equiv 3 \pmod 4$. The unique factor of $\prod_{k=0}^{p-1}(4k+3)$ divisible by p is $4j+3 = rp$. All remaining factors are p-adic units. Writing $k = j+t$ and cancelling p against $p!$ yields the exact identity

$$a(p) = 2^p r\, 4^{p-1} \frac{(-1)^j}{\binom{p-1}{j}} \prod_{\substack{-j \le t \le p-1-j \\ t \neq 0}} \left(1 + \frac{rp}{4t}\right).$$

As $\binom{p-1}{j} = (-1)^j \prod_{i=1}^{j}(1 - p/i)$, the factorial ratio is $1 + pH_j$ modulo p^2. Also

$$\prod_{t \neq 0} \left(1 + \frac{rp}{4t}\right) \equiv 1 + \frac{rp}{4}\,(H_{p-1-j} - H_j) \equiv 1 \pmod{p^2}$$

by Lemma 1. Therefore

$$a(p) \equiv 2r\,(2^{p-1})^3 (1 + pH_j) \equiv 2r\big(1 + p(3q + H_j)\big) \pmod{p^2}. \tag{1}$$

If $p \equiv 3 \pmod 4$, then $j = \lfloor p/4 \rfloor$. If $p \equiv 1 \pmod 4$, then $p - 1 - j = \lfloor p/4 \rfloor$. Lemma 1 shows in both cases that $H_j \equiv -3q \pmod p$. The parenthesis in (1) is therefore 1 modulo p^2, proving $a(p) \equiv 2r \pmod{p^2}$. □

Scope and checks

For $p = 3, 5, 7, 13, 17, 19$ the residues $a(p) \bmod p^2$ are respectively $2, 6, 2, 6, 6, 2$, in agreement with the theorem. Since $6 \not\equiv 2 \pmod{p^2}$ for any odd p, this also shows the $p \equiv 1 \pmod 4$ condition in the OEIS conjecture cannot simply be removed. The result treats $n = 1$ only, not arbitrary n.

AI authorship and verification disclosure. Written by ChatGPT (OpenAI); the algebraic derivation and sample residues were checked in this run, and the PDF was rendered and inspected. No independent human peer review or claim of mathematical novelty is made.

References

[1] OEIS Foundation, *A004982: Hypergeometric product sequence with generating function* $(1-8x)^{-3/4}$, https://oeis.org/A004982, accessed 14 November 2026.

[2] *On a congruence of Emma Lehmer*, https://www.impan.pl/shop/en/publication/transaction/download/product/82378, section 7 (context for the classical quarter-harmonic congruence, which is also proved above).